

RADIATION AND MASS TRANSFER EFFECTS ON FREE CONVECTION FLOW OF A DISSIPATIVE FLUID THROUGH A HIGHLY POROUS MEDIUM

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The unsteady natural convection flow of a viscous, incompressible and dissipative fluid through a porous medium with high porosity bounded by a vertical infinite stationary plate in the presence of radiation has been investigated. The fluid is gray, absorbing-emitting but non-scattering medium and the Rosseland approximation is used to describe the radiative flux in the energy equation. Approximate solutions have been obtained for velocity, temperature and concentration fields using multiparameter perturbation technique. The effects of various parameters have been discussed and shown graphically on velocity, temperature and concentration.

INTRODUCTION

Transport of momentum and thermal energy in fluid-saturated porous media with low porosities is commonly described by Darcy's model for conservation of momentum and by the energy equation based on the velocity field found from this model (Kaviany [1]). In contrast to rocks, soil, sand and other media that do fall in this category, certain porous materials, such as foam metals and fibrous media, usually have high porosity. Vajravelu [2] examined the steady flow and heat transfer in a porous medium with high porosity. Hong *et al* [3], Chen and Lin [4] and Jaiswal and Soundalgekar [5] studied the natural convection in a porous medium with high porosity.

For some industrial applications such as glass production and furnace design, and in space technology applications such as cosmical flight aerodynamics rocket, propulsion systems, plasma physics and space craft re-entry aerothermodynamics which operate at higher temperatures, radiation effects can be significant. Chandrasekhara and Nagaraju [6] examined the composite heat-transfer in a variable porosity medium bounded by an infinite flat-plate in the presence of radiation. Raptis and Perdakis [7] studied unsteady flow through a porous medium with high porosity bounded by a vertical infinite plate in the presence of radiation.

In all the studies mentioned above, viscous dissipation is neglected. But the viscous dissipation in the natural convective flow is important, when the flow field is of extreme size or in high gravitational field. In view of this, Ramachandra Prasad and Bhaskar Reddy [8] investigated the radiation and mass transfer effects on an unsteady MHD free convection flow past a heated vertical plate in a porous medium with viscous dissipation. The object of this paper is to study the radiation effects on an unsteady natural convection flow of a viscous incompressible and dissipative fluid through a porous medium with high porosity bounded by a vertical infinite stationary plate with mass transfer.

MATHEMATICAL ANALYSIS

An unsteady two-dimensional laminar free convection flow through a porous medium with high porosity is considered. The porous medium is bounded by a vertical infinite stationary plate. The fluid and the porous structure are assumed to be in local thermal equilibrium. It is also assumed that the plate temperature and concentration near the plate vary exponentially with time. The x' -axis is chosen along the plate with direction opposite to the direction of gravity and the y' -axis is taken normal to the plate. The radiative heat flux in the x' -direction is assumed to be negligible in comparison to that in the y' -direction. Now, under the usual Boussinesq's approximation, the equations of continuity, momentum, energy and species, which govern the flow field are

$$\frac{\partial v'}{\partial y'} = 0 \quad \dots (2.1)$$

$$\rho \left(\frac{\partial u'}{\partial t'} + v' \frac{\partial u'}{\partial y'} \right) = -\frac{\partial P}{\partial x'} - \rho g + \nu \rho \frac{\partial^2 u'}{\partial y'^2} - \frac{\nu \rho}{K'} \phi u' \quad \dots (2.2)$$

$$\sigma \frac{\partial T'}{\partial t'} + \phi v' \frac{\partial T'}{\partial y'} = \frac{k_m}{\rho c_p} \frac{\partial^2 T'}{\partial y'^2} - \frac{\phi}{\rho c_p} \frac{\partial q_r}{\partial y'} + \frac{\nu}{c_p} \left(\frac{\partial u'}{\partial y'} \right)^2 \quad \dots (2.3)$$

$$\frac{\partial C'}{\partial t'} + v' \frac{\partial C'}{\partial y'} = D \frac{\partial^2 C'}{\partial y'^2} \quad \dots (2.4)$$

The boundary conditions for the velocity, temperature and concentration fields are

$$u' = 0, T' = T_w' + \varepsilon \left(T_w' - T_\infty' \right) e^{n' t'}, C' = C_w' + \varepsilon \left(C_w' - C_\infty' \right) e^{n' t'} \quad \text{at } y' = 0$$

$$u' \rightarrow U_\infty', T' \rightarrow T_\infty', C' \rightarrow C_\infty', \quad \text{as } y' \rightarrow \infty \quad \dots (2.5)$$

where all the physical variables have their usual meaning.

The suction velocity normal to the plate is taken as $v' = -v_0$

By using the Rosseland approximation (Brewster [9]), the radiative heat flux term is taken as

$$q_r = -\frac{4\sigma_s}{3k_e} \frac{\partial T'^4}{\partial y'} \quad \dots (2.6)$$

where σ_s is the Stefan-Boltzmann constant and k_e the mean absorption coefficient.

In order to write the governing equations and the boundary conditions in dimensionless form, the following non-dimensional quantities are introduced.

$$u = \frac{u'}{U_\infty}, y = \frac{y'}{v_0}, t = \frac{t' v_0^2}{\nu}, n = \frac{\nu n'}{v_0^2}, \lambda = \frac{\sigma}{\phi}, T = \frac{T' - T_\infty'}{T_w' - T_\infty'}, C = \frac{C' - C_\infty'}{C_w' - C_\infty'}$$

$$Pr = \frac{\phi \rho v C_p}{k_m}, \quad Ec = \frac{V_0^2}{C_p (T_w' - T_\infty')}, \quad K = \frac{K' v_0^2}{\phi v^2},$$

$$Gr = \frac{vg\beta (T_w' - T_\infty')}{U_\infty'^2}, \quad Gm = \frac{vg\beta^* (C_w' - C_\infty')}{U_\infty'^2}, \quad N = \frac{k_e k_m}{4\phi\sigma_s T_\infty'^{+3}} \quad \dots (2.7)$$

In view of Equation (2.13), Equations (2.9), (2.12) and (2.4) reduce to the following dimensionless form

$$\frac{\partial u}{\partial t} - \frac{\partial u}{\partial y} = GrT + GmC + \frac{\partial^2 u}{\partial y^2} + \frac{1}{K}(1-u) \quad \dots (2.8)$$

$$\lambda \frac{\partial T}{\partial t} - \frac{\partial T}{\partial y} = \frac{1}{Pr} \left(1 + \frac{4}{3N} \right) \frac{\partial^2 T}{\partial y^2} + Ec \left(\frac{\partial u}{\partial y} \right)^2 \quad \dots (2.9)$$

$$\frac{\partial C}{\partial t} - \frac{\partial C}{\partial y} = \frac{1}{Sc} \frac{\partial^2 C}{\partial y^2} \quad \dots (2.10)$$

where Pr , Ec , K , Gr , Gm and N are the effective Prandtl number, Eckert number, permeability parameter, thermal Grashof number, species Grashof number and radiation parameter respectively.

The corresponding boundary conditions are

$$\begin{aligned} u = 0, \quad T = 1 + \varepsilon e^{\eta}, \quad C = 1 + \varepsilon e^{\eta} & \quad \text{at } y = 0, \\ u \rightarrow 1, \quad T \rightarrow 0, \quad C \rightarrow 0 & \quad \text{as } y \rightarrow \infty \end{aligned} \quad \dots (2.13)$$

SOLUTION OF THE PROBLEM

The equations (2.8) - (2.10) are coupled non-linear partial differential equations and these cannot be solved in closed form. To linearize these equations, it is assumed that the unsteady flow is superimposed on the mean steady flow. Then, in the neighborhood of the plate, it is considered that

$$u(y, t) = u_0(y) + \varepsilon u_1(y) e^{\eta}, \quad T(y, t) = T_0(y) + \varepsilon T_1(y) e^{\eta}, \quad C(y, t) = C_0(y) + \varepsilon C_1(y) e^{\eta} \quad \dots (3.1)$$

Substituting Equation (3.1) into the system of Equations (2.8) - (2.10) and equating the harmonic and non-harmonic terms, and neglecting the higher order terms of $O(\varepsilon)^2$, we get

$$u_0'' + u_0' - \frac{1}{K} u_0 = -\frac{1}{K} GrT_0 - GmC_0 \quad \dots (3.2)$$

$$u_1'' + u_1' - \left(\frac{1}{K} + n \right) u_1 = -GrT_1 - GmC_1 \quad \dots (3.3)$$

$$\frac{1}{Pr} \left[1 + \frac{4}{3N} \right] T_0'' + T_0' = -Ec \left(U_0' \right)^2 \quad \dots (3.4)$$

$$-\frac{1}{Pr} \left[1 + \frac{4}{3N} \right] T_1'' + T_1' - \lambda n T_1 = -2Ec u_0' u_1' \quad \dots (3.5)$$

$$C_0'' + Sc C_0' = 0 \quad \dots (3.6)$$

$$C_1'' + Sc C_1' - n Sc C_1 = 0 \quad \dots (3.7)$$

The corresponding boundary conditions can be written as

$$u_0 = 0, u_1 = 0, T_0 = 1, T_1 = 1, C_0 = 1, C_1 = 1 \quad \text{at} \quad y = 0 \quad \dots (3.8)$$

$$u_0 \rightarrow 1, u_1 \rightarrow 0, T_0 \rightarrow 0, T_1 \rightarrow 0, C_0 \rightarrow 0, C_1 \rightarrow 0 \quad \text{as} \quad y \rightarrow \infty$$

By solving the differential equations (3.6) and (3.7) under the boundary conditions (3.8) and substituting the solutions into Equation (3.1), we get the analytic solution for the concentration field in terms of y and t as

$$C = e^{-Sc \cdot y} + \varepsilon e^{\mu t} e^{-B_1 \cdot y} \quad \text{where, } B_1 = \frac{Sc + \sqrt{Sc^2 + 4Scn}}{2} \quad \dots (3.9)$$

The equations (3.2)–(3.5) are still coupled and non-linear, whose exact solutions are not possible. So, we expand u_0, u_1, T_0, T_1 in terms of Ec in the following form, as the Eckert number is very small for incompressible flows.

$$\begin{aligned} u_0(y) &= u_{01}(y) + Ec u_{02}(y), \quad u_1(y) = u_{11}(y) + Ec u_{12}(y) \\ T_0(y) &= T_{01}(y) + Ec T_{02}(y), \quad T_1(y) = T_{11}(y) + Ec T_{12}(y) \end{aligned} \quad \dots (3.10)$$

Substituting (3.10) in Equations (3.2)–(3.5), equating the coefficients of Ec to zero and neglecting the terms in Ec^2 and higher order, we get the zeroth and first order equations. Solving these equations with their respective boundary conditions and using Equations (3.10) and (3.1), we obtain the velocity and temperature distributions in the boundary layer as

$$\begin{aligned} u(y, t) &= (S_2 + S_3 - 1) e^{-B_2 y} - S_2 e^{-B_2 y} - S_3 e^{-S_3 y} + 1 \\ &\quad + Ec \{ A_{13} e^{-B_2 y} - S_{13} e^{-B_2 y} + S_{14} e^{-2B_2 y} + S_{15} e^{-2B_2 y} + S_{16} e^{-2S_3 y} - S_{17} e^{-S_3 y} \} \\ &\quad + \varepsilon e^{\mu t} \{ \{ A_{14} e^{-B_1 y} - S_{18} e^{-B_1 y} - S_{19} e^{-B_1 y} \} \\ &\quad + Ec \{ A_{18} e^{-B_1 y} - S_{38} e^{-B_2 y} + S_{39} e^{-S_{20} y} - S_{40} e^{-S_{21} y} - S_{41} e^{-S_{22} y} - S_{42} e^{-S_{23} y} + S_{43} e^{-S_{24} y} \\ &\quad + S_{44} e^{-S_{25} y} - S_{45} e^{-S_{26} y} + S_{46} e^{-S_{27} y} - S_{47} e^{-S_{28} y} \} \\ &\quad + S_{48} e^{-S_{29} y} - S_{49} e^{-S_{30} y} + S_{50} e^{-S_{31} y} + S_{51} e^{-S_{32} y} + S_{52} e^{-S_{33} y} \\ &\quad + S_{53} e^{-S_{34} y} - S_{54} e^{-S_{35} y} + S_{55} e^{-S_{36} y} - S_{56} e^{-S_{37} y} - S_{57} e^{-S_{38} y} \} \\ &\quad + \varepsilon e^{\mu t} \{ \{ e^{-B_2 y} \} + Ec \{ A_{16} e^{-B_2 y} - S_{29} e^{-S_{20} y} + S_{30} e^{-S_{21} y} + S_{31} e^{-S_{22} y} \} \} \end{aligned}$$

Knowing the velocity field, the skin-friction at the plate can be obtained, which in the non-dimensional form is given by,

$$\tau = \left[\frac{\partial u}{\partial y} \right]_{y=0} = \left[\frac{\partial u_0}{\partial y} + \varepsilon e^{\mu t} \frac{\partial u_1}{\partial y} \right]$$

Knowing the temperature field, the rate of heat transfer coefficient can be obtained, which in the non-dimensional form, in terms of Nusselt number, is given by

$$Nu = - \left[\frac{\partial T}{\partial y} \right]_{y=0} = - \left[\frac{\partial T_0}{\partial y} + \epsilon e^{\mu} \frac{\partial T_1}{\partial y} \right]_{y=0}$$

Knowing the concentration field the rate of mass transfer coefficient can be obtained, which in the non-dimensional form, in terms of the Sherwood number, is given by

$$Sh = \left[\frac{\partial C}{\partial y} \right]_{y=0} = - \left[\frac{\partial C_0}{\partial y} + \epsilon e^{\mu} \frac{\partial C_1}{\partial y} \right]_{y=0}$$

RESULTS AND DISCUSSION

In order to get a physical insight of the problem, the above physical quantities are computed numerically for different values of the governing parameters viz., thermal Grashof number Gr , the solutal Grashof number Gm , Prandtl number Pr , Schmidt number Sc , the radiation parameter N , the permeability of the porous medium K , and Eckert number Ec and are presented in figures and tables.

It is observed from Fig. 1 that an increase in Gr , leads to a rise in the values of velocity due to the enhancement in buoyancy force. For the case of different values of the solutal Grashof number, the velocity profiles in the boundary layer are shown in Fig. 2. The velocity distribution attains a distinctive maximum value in the vicinity of the plate and then decreases properly to approach a free stream value. It is noticed that an increase in the radiation parameter results in a decrease in the velocity and temperature with in the boundary layer, as well as decreased thickness of the velocity and temperature boundary layers are shown in Figs. 3(a) and 3(b).

Figs. 4(a) and 4(b) display the effects of Schmidt number on the velocity and concentration respectively. As the Schmidt number increases, the velocity and concentration decreases. The effects of the viscous dissipation parameter *i.e.*, the Eckert number on the velocity and temperature are shown in Figs. 5(a) and 5(b). Greater viscous dissipative heat causes a rise in the velocity as well as temperature.

Figs. 6(a) and 6(b) illustrate the behavior of the velocity and temperature for different values of the Prandtl number. The numerical results show that the effect of increasing values of Prandtl number results in a decreasing velocity. From Fig. 6 (b), it is observed that an increase in the Prandtl number results in a decrease of the thermal boundary layer thickness and in general lower average temperature with in the boundary layer. Fig. 7 shows the velocity profiles for different values of the permeability parameter. Clearly, as K increases the peak values of the velocity tends to increase. Figs. 8 and 9 depicts the temperature profiles for different buoyancy parameters Gr or Gm . It is observed that, owing to an increase in the values of the buoyancy parameters Gr or Gm , the temperature decreases.

Tables 1-5 show the effects of the thermal Grashof number, solutal Grashof number, radiation parameter, Schmidt number and Eckert number on the skin friction τ , Nusselt number Nu , and Sherwood number Sh . From tables 1 and 2, it is observed that, as Gr or Gm increases, the skin-friction coefficient increases. From table 3, it can be seen that, as the radiation parameter increases, the skin-friction decreases, where as the Nusselt number increases. From table 4, it is noted that an increase in the Schmidt number reduces both the skin-friction and Sherwood number. From table 5, it is observe that as the Eckert number increases the skin-friction increases, whereas the Nusselt number decreases.

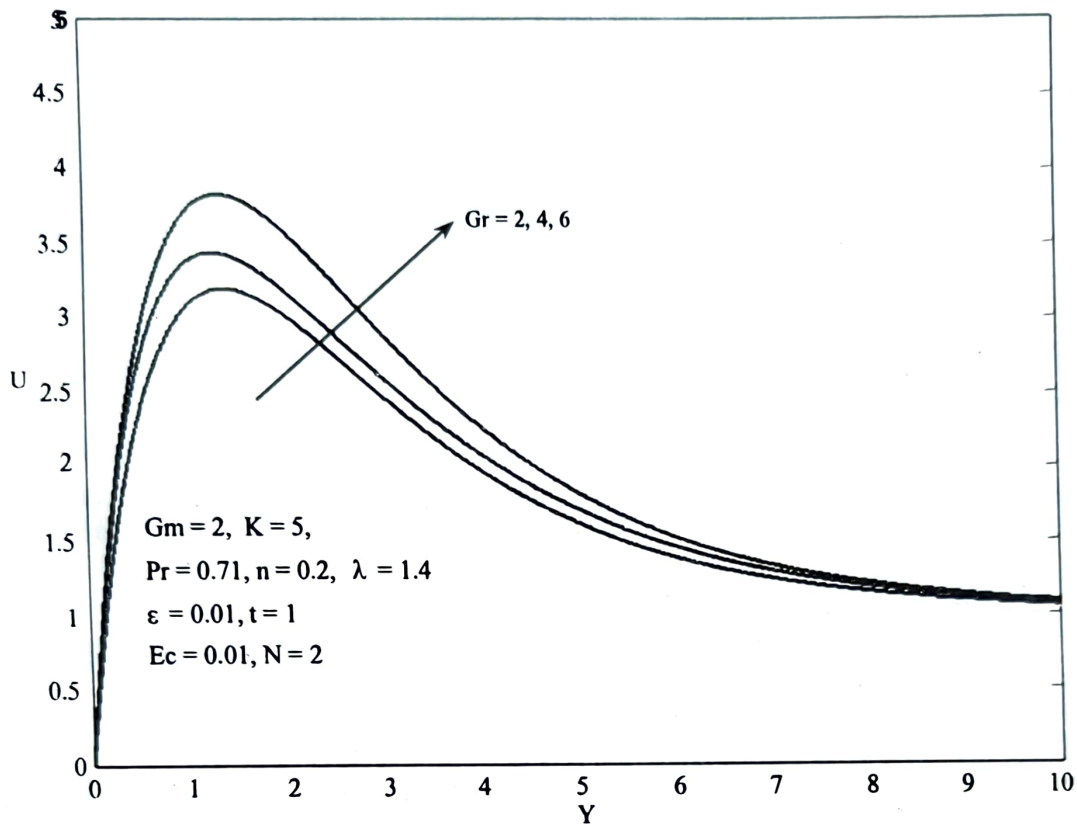


Fig. 1. Velocity profiles for different values of Gr.

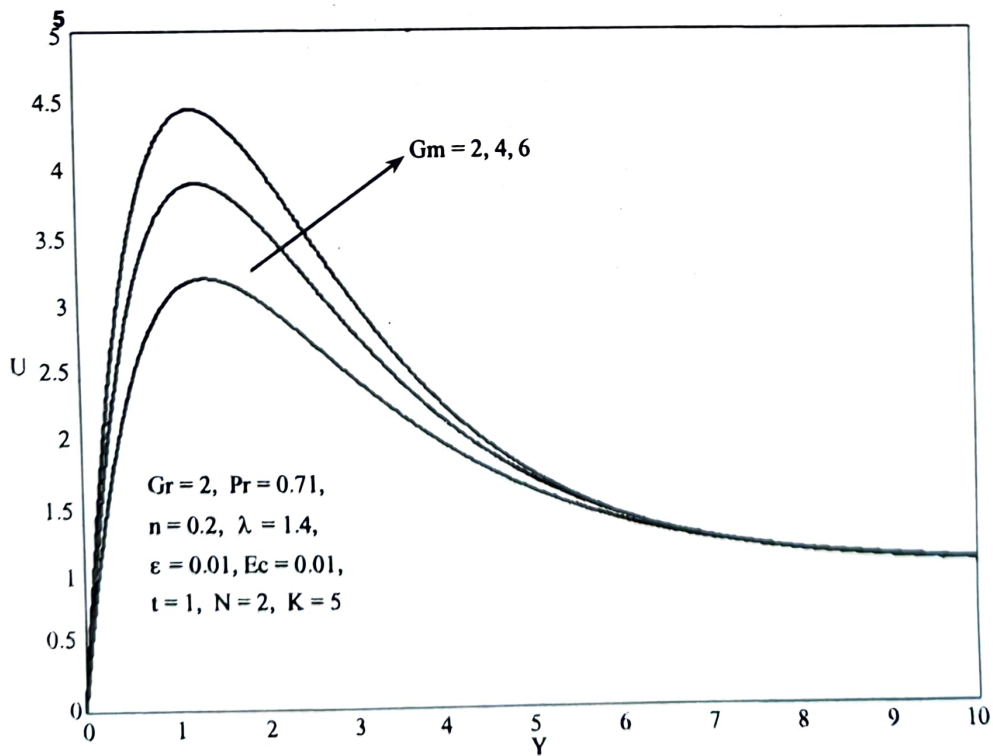


Fig. 2. Velocity profiles for different values of Gm

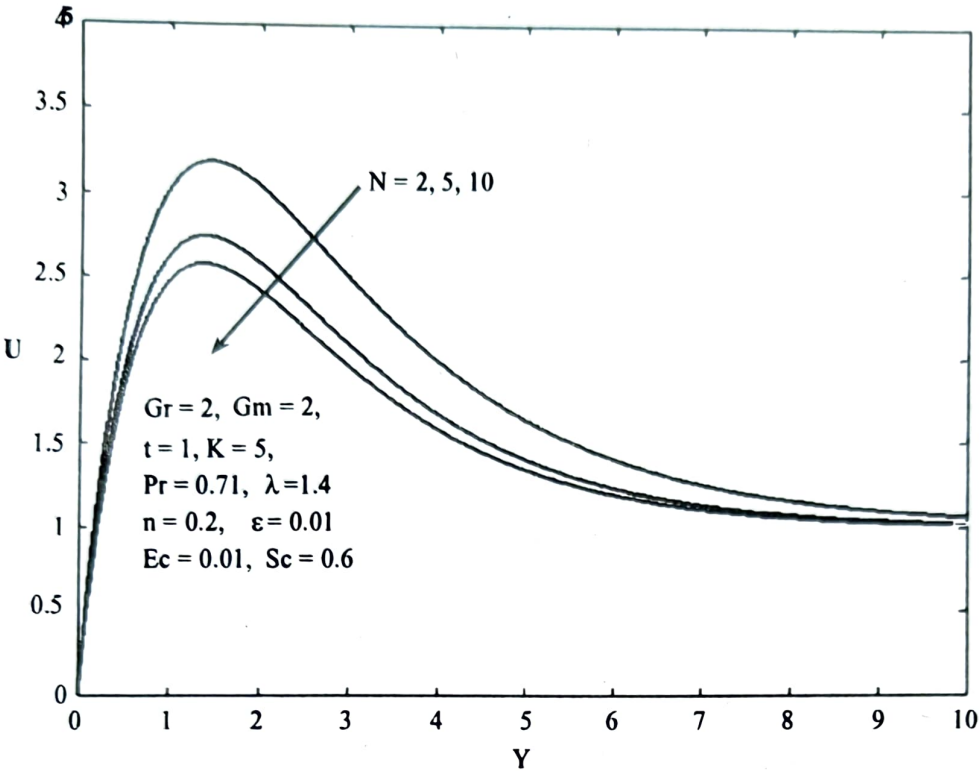


Fig. 3(a). Velocity profiles for different values of N

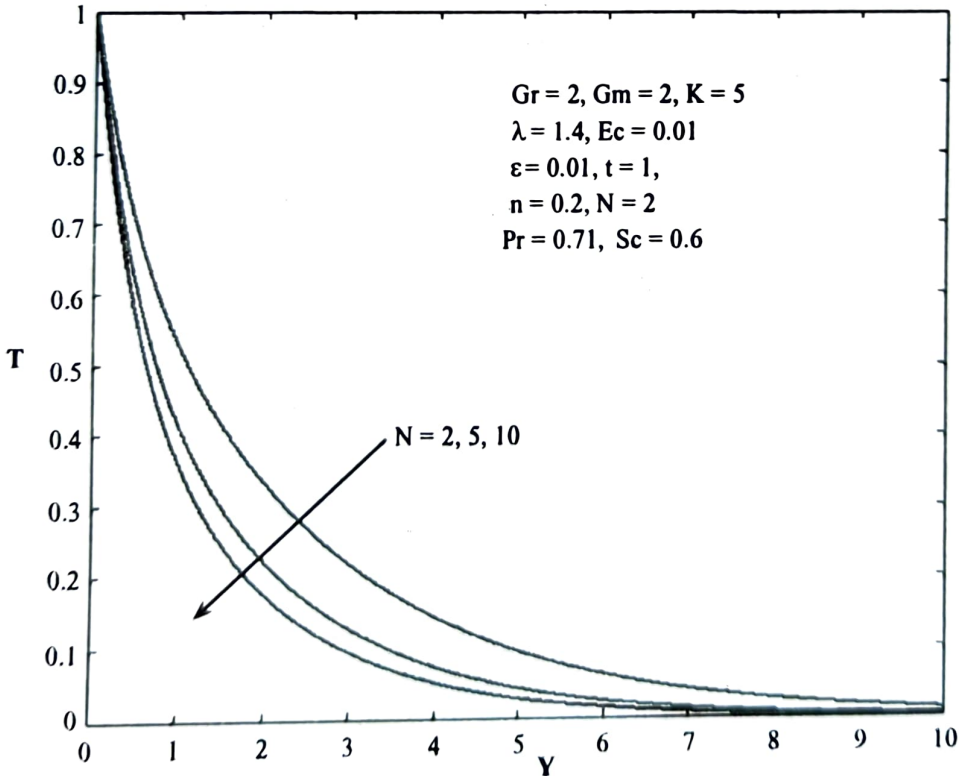


Fig. 3(b). Temperature profiles for different values of N

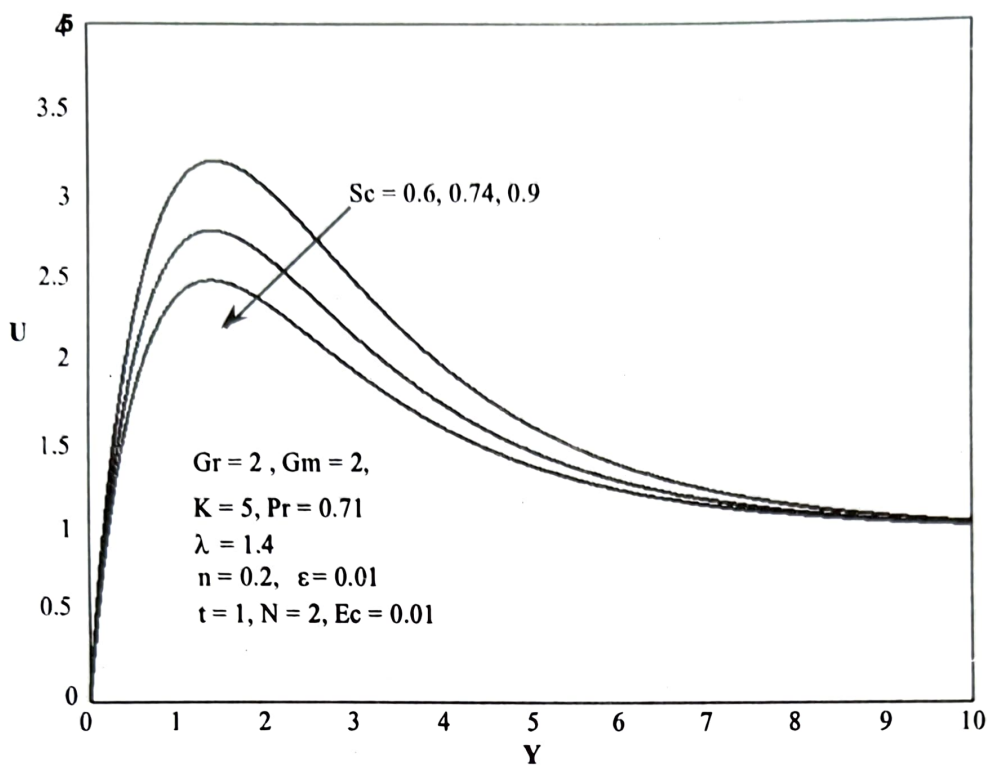


Fig. 4 (a). Velocity profiles for different values of Sc

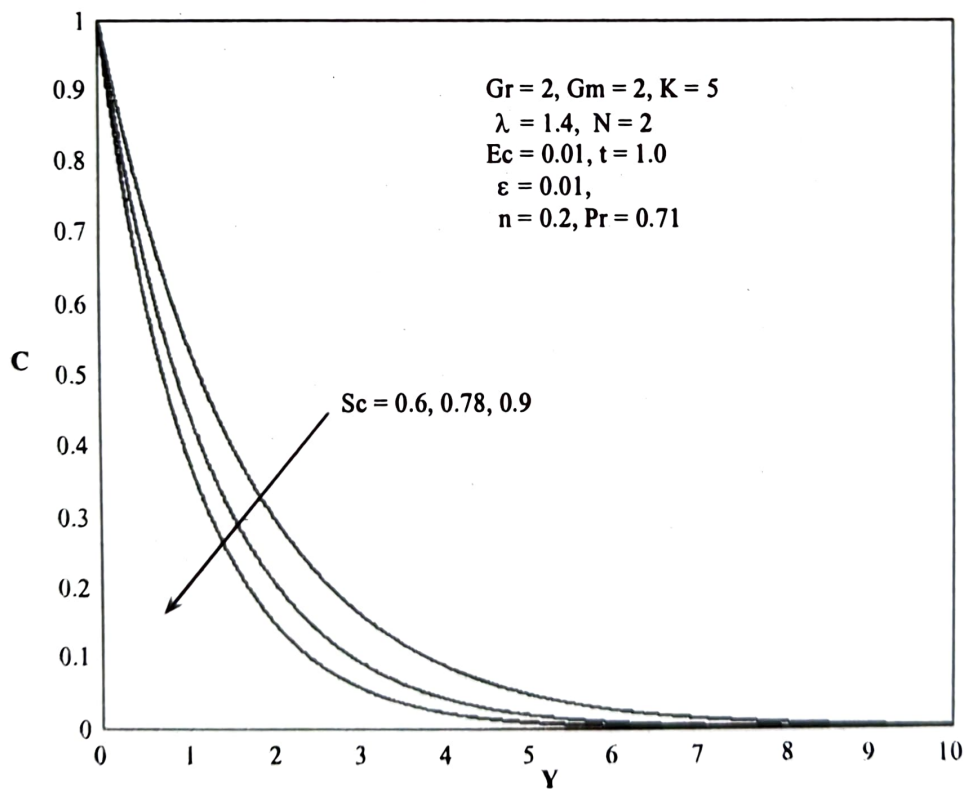


Fig. 4 (b). Concentration profiles for different values of Sc

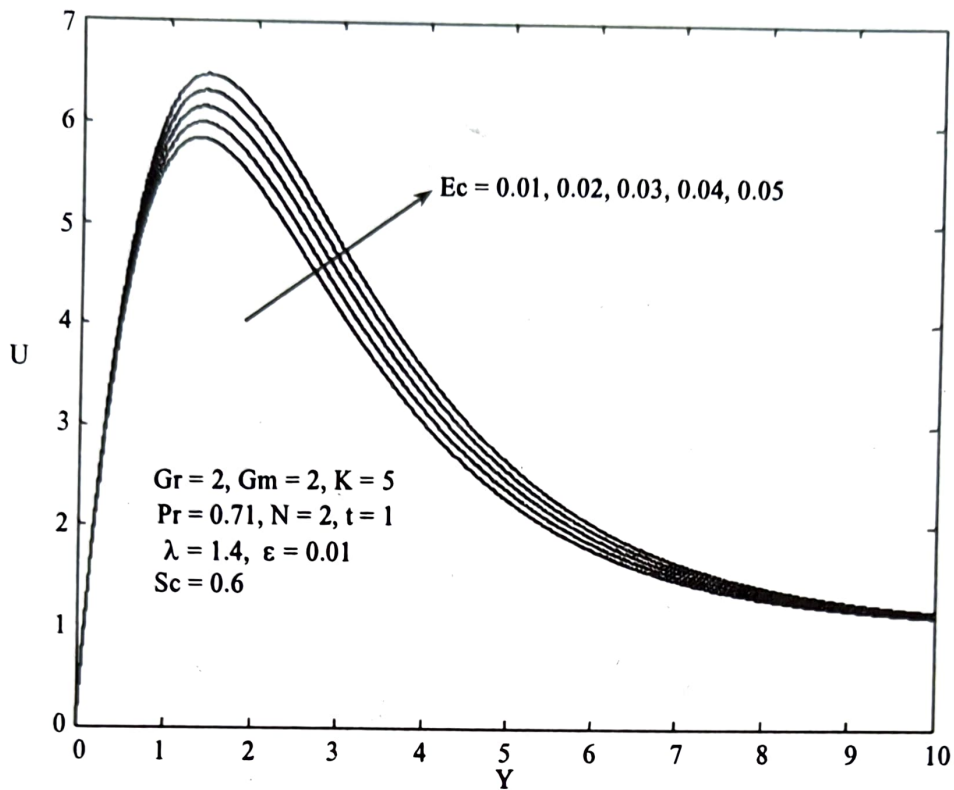


Fig. 5(a). Velocity profiles for different values of Ec

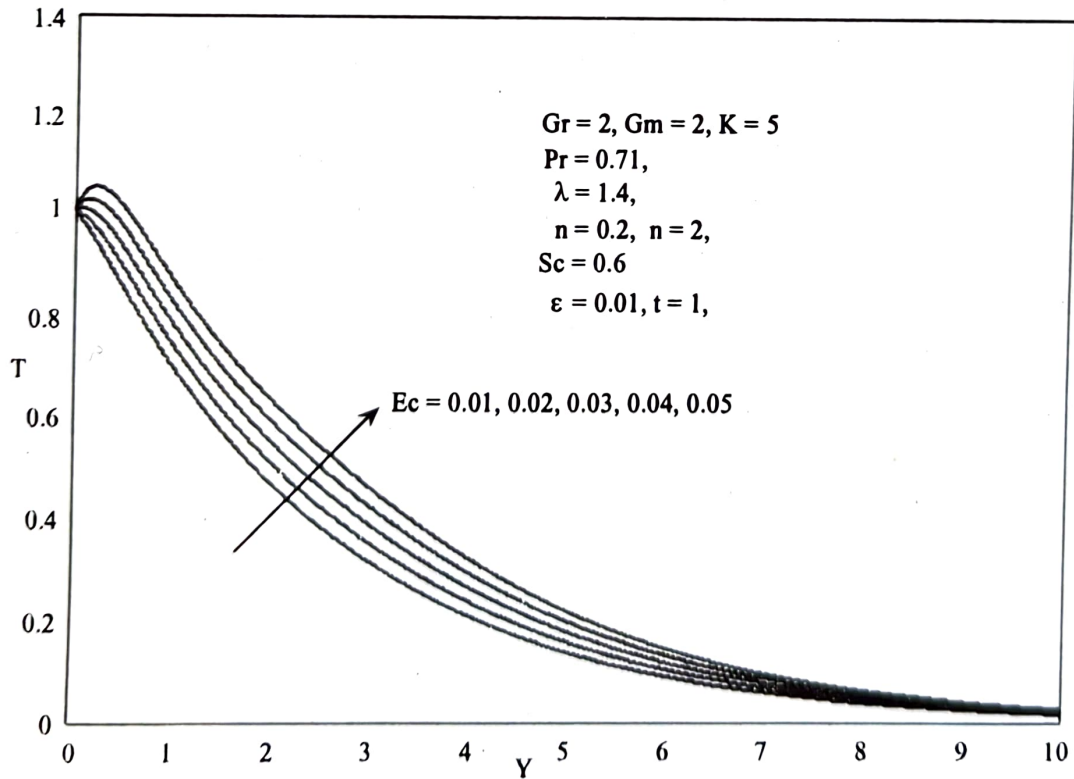


Fig. 5(b) Temperature profiles for different values of Ec

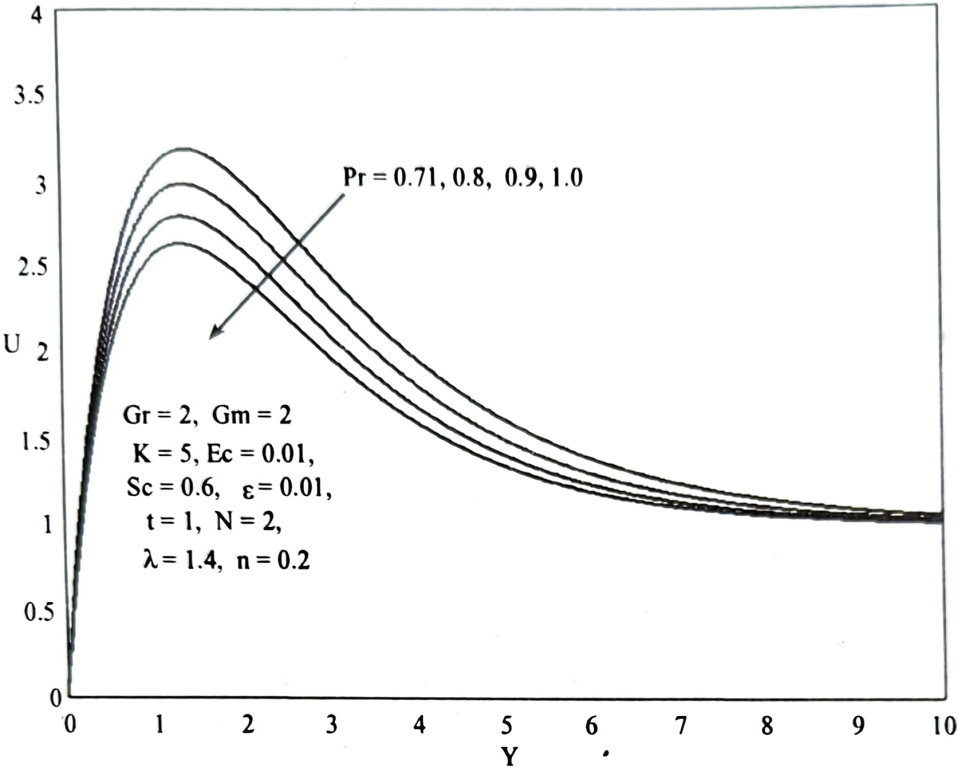


Fig. 6 (a) Velocity profiles for different values of Pr

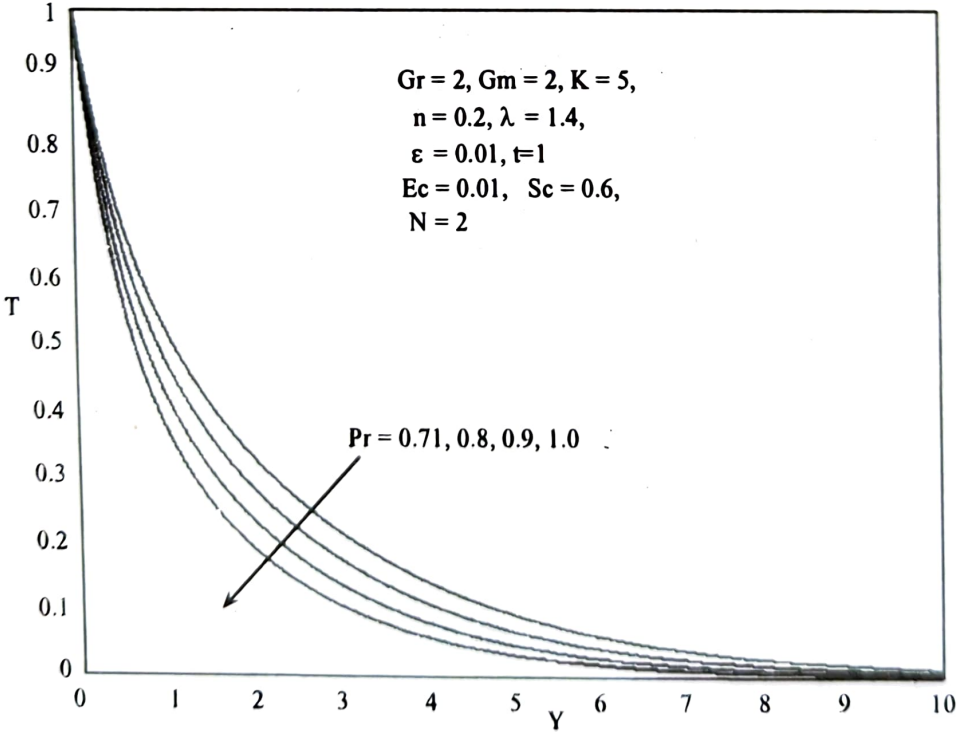


Fig. 6(b) Temperature profiles for different values of Pr